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To the Editors-in-Chief, *Software and Systems Modeling*

Re: Submission of “Behavioral Authority in Sealed Models: Execution Semantics and Governed Evolution”

Dear Editors,

Please consider this manuscript for publication in *Software and Systems Modeling*. It addresses a basic question in model-driven engineering: can an executable model be the authoritative source of domain-visible behavioral decisions, so that the tools that build and run the system do not add decisions the model never made?

The paper defines six kinds of domain-visible decisions and presents an execution semantics for a *sealed behavioral model*. The runtime follows only routes and effects declared in that model; when a required decision is missing, it refuses rather than filling the gap. The paper also states three properties for a conforming implementation and turns them into checks for construction, execution, and recorded traces. It describes a supporting approach to model change in which each new version is derived from a named, unchanged predecessor.

We evaluate one open reference implementation, Protocol-Governed Computing, across four application domains. The tests examine whether decisions can be traced to the model, whether altered model snapshots are rejected, how the implementation handles omissions, and whether captured inputs support replay. On the exercised paths of 243 traces, all 3,019 recorded decisions were attributable to model declarations, and acceptance rejected all nine tested snapshot alterations. The evaluation also reports important failures: construction accepted a model with a missing route, which execution refused only later; in another case, the runtime continued past an outcome omitted by a step, adding a decision without recording it. These findings show both the promise of the approach and the value of checks that can expose nonconformance. The evidence comes from one implementation family and covers only some of the six decision classes; the manuscript states these limits and distinguishes its normative semantics from its empirical results.

We believe the paper fits the journal’s interest in modeling languages, executable models, and their semantics. It engages with related work on models at run time, model transformation, and traceability published in *Software and Systems Modeling*.

I am an independent researcher. The work received no funding, and I declare no competing interests. Section 1.5 identifies five related works by the author, including one under review elsewhere; that work addresses a different research question and shares no experiment or data with this submission. The manuscript’s Declarations section also describes the author’s

use of generative AI tools. The evaluated composition and supplementary experiment materials are available at <https://doi.org/10.5281/zenodo.23129879>.

This manuscript is original, has not been published previously, and is not under consideration elsewhere.

Thank you for your consideration.

Sincerely,
Bhash Ganti